BMAS 1103: ENGINEERING MATHEMATICS III

Course Objectives: To make the students understand the concepts of Statistics, Laplace transform and Complex analysis by giving more emphasis to their applications in engineering.

Credits: 04 **Semester III** **L–T–P: 3–1–0**

Module No.	Contents	Teaching Hours (Approx.)
I	**Statistics:** Introduction, Fitting of straight line by the method of least squares, Correlation, Regression lines, Binomial and Poisson distributions, Sampling, Level of significance, Chi-square test. **Laplace Transform**: Properties of Laplace transform, Laplace transform of derivatives and integrals, Laplace transform of periodic functions, Properties of inverse Laplace transform, convolution theorem, Engineering applications in solving ordinary differential equations.	20
II	**Complex Analysis**: Analytic functions, Harmonic functions, Harmonic conjugates, Milne's Thomson method, Conformal mapping, Bilinear transformation, Line integral in a complex plane, Cauchy's integral theorem, Cauchy's integral formula, Taylor and Laurent series (without proof), Singularities, Residue at a pole, Cauchy's residue theorem and its application in evaluation of real integrals (excluding poles on real axis).	20

Course Outcomes:

After studying these topics, the student will be able to

- **CO 1:** Compute the coefficient of correlation and regression between two variables
- **CO 2:** Apply Chi-square test
- **CO 3:** Use Laplace transform in solving differential equations
- **CO 4:** Know the concept of analytic functions and its applications in engineering
- **CO 5:** Understand bilinear transformation and its applications
- **CO 6:** Use Cauchy's residue theorem in solving real integrals

Text Books:

- M. Goyal and N. P. Bali, A Text Book of Engineering Mathematics, Laxmi Publication, Delhi, 2014.
- B. S. Grewal, Higher Engineering Mathematics, Khanna Publishers, New Delhi, 2015.

Reference Books:

- R. V. Churchill and J. W. Brown, Complex Variables and Applications, McGraw-Hill, New York, 2013.
- S. C. Gupta and V. K. Kapoor, Fundamentals of Mathematical Statistics, Sultan Chand Publications, Delhi, 2014.
- M. K. Jain, S. R. K. Iyengar and R. K. Jain, Advanced Engineering Mathematics, Narosa Publishing House, New Delhi, 2002.